Fluid Mechanics: Dyads

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Constant density Navier-Stokes Equations

Momentum $$\frac{D\vec{u}}{Dt} = -\frac{1}{\rho_0}\nabla p + \nu\nabla^2\vec{u}$$

Mass $$\vec{\nabla}\cdot\vec{u} = 0$$

These equations are written in coordinate free form.

They are also written so that all but the terms in red can be expressed in terms of standard vector operators.

The terms in red are the VECTOR Laplacian.

- The idea of writing the equations in a coordinate free form is vital for a physically valid description of a continuum.
- To understand how a particular coordinate system that is not a simple Cartesian coordinate system is used takes a bit of work.
- There are two choices.
- We could use a very general differential geometric point of view and then express the various tensors in a particular coordinate system using the Christoffel symbols (https://en.wikipedia.org/wiki/Christoffel_symbols).
- When the coordinate system is orthogonal, as is the case for spherical and cylindrical coordinates dyads are a far simpler, and physically more meaningful choice.

- Recall Cauchy's theorem for expressing the traction:

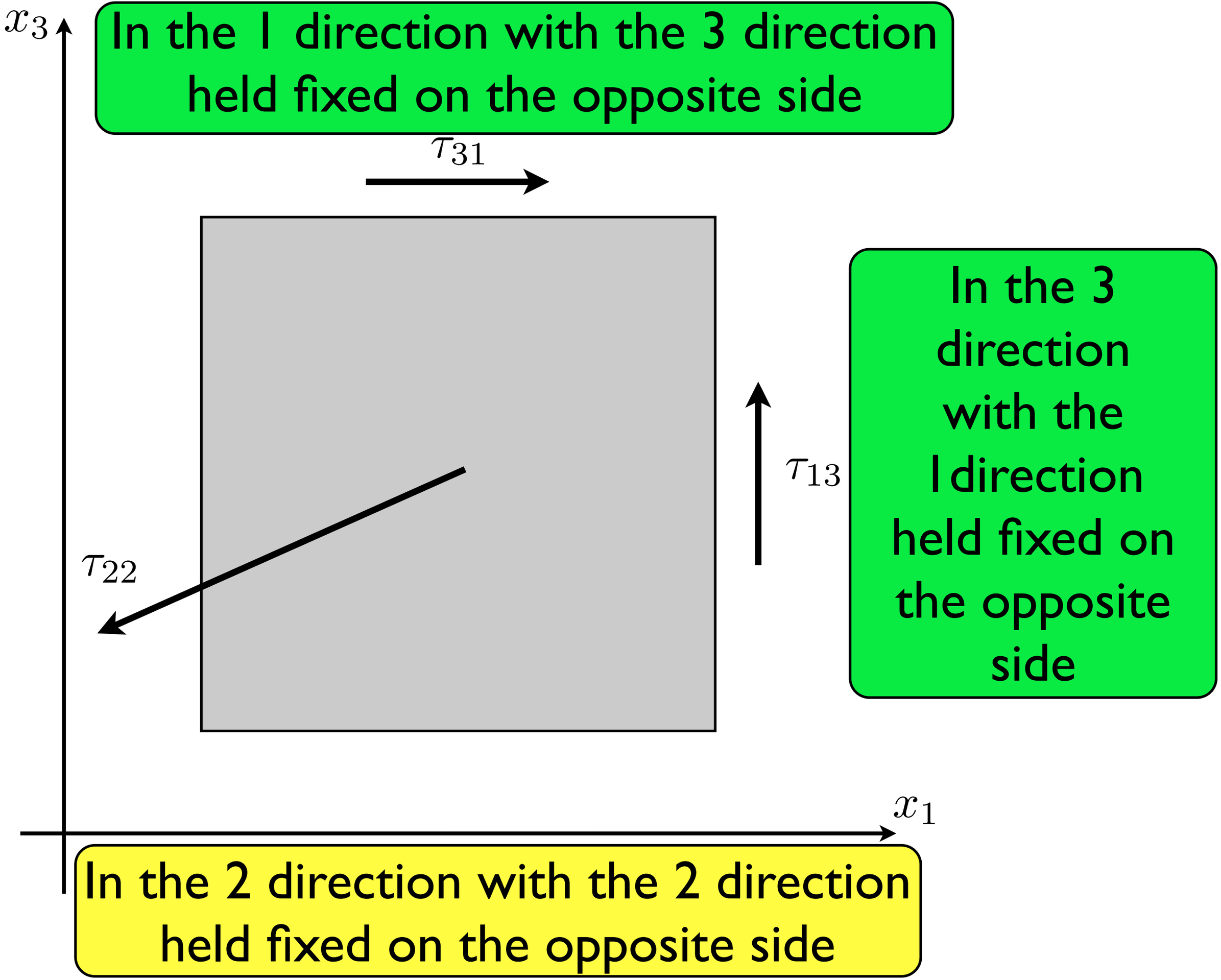
$$\vec{t}(\vec{x};\hat{n}) = \tau_{ij} n_j$$

- The right hand side is a contraction between a second order tensor called stress and the unit normal that defines the "inside" and "outside" of an orientable surface.

- The same way we use the basis vectors of a coordinate system to write a vector we could ask "What is the basis for the stress tensor"

- In almost any source on continuum mechanics we learn that the standard way to interpret stress is as the forces on a cube.

- On each face we can act in one normal direction to the face, and two shear directions (much easier to visualize for a square in the plane).


In the 1 direction with the 3 direction held fixed on the opposite side
x_3
τ_{31}
τ_{13}
In the 3 direction with the 1 direction held fixed on the opposite side
τ_{22}
x_1
In the 2 direction with the 2 direction held fixed on the opposite side

- While a vector is written in terms of a weighted sum of the individual basis vectors the stress tensor is written as the weighted sum of the pairwise outer product of the basis vectors.

- The latter are called dyads.

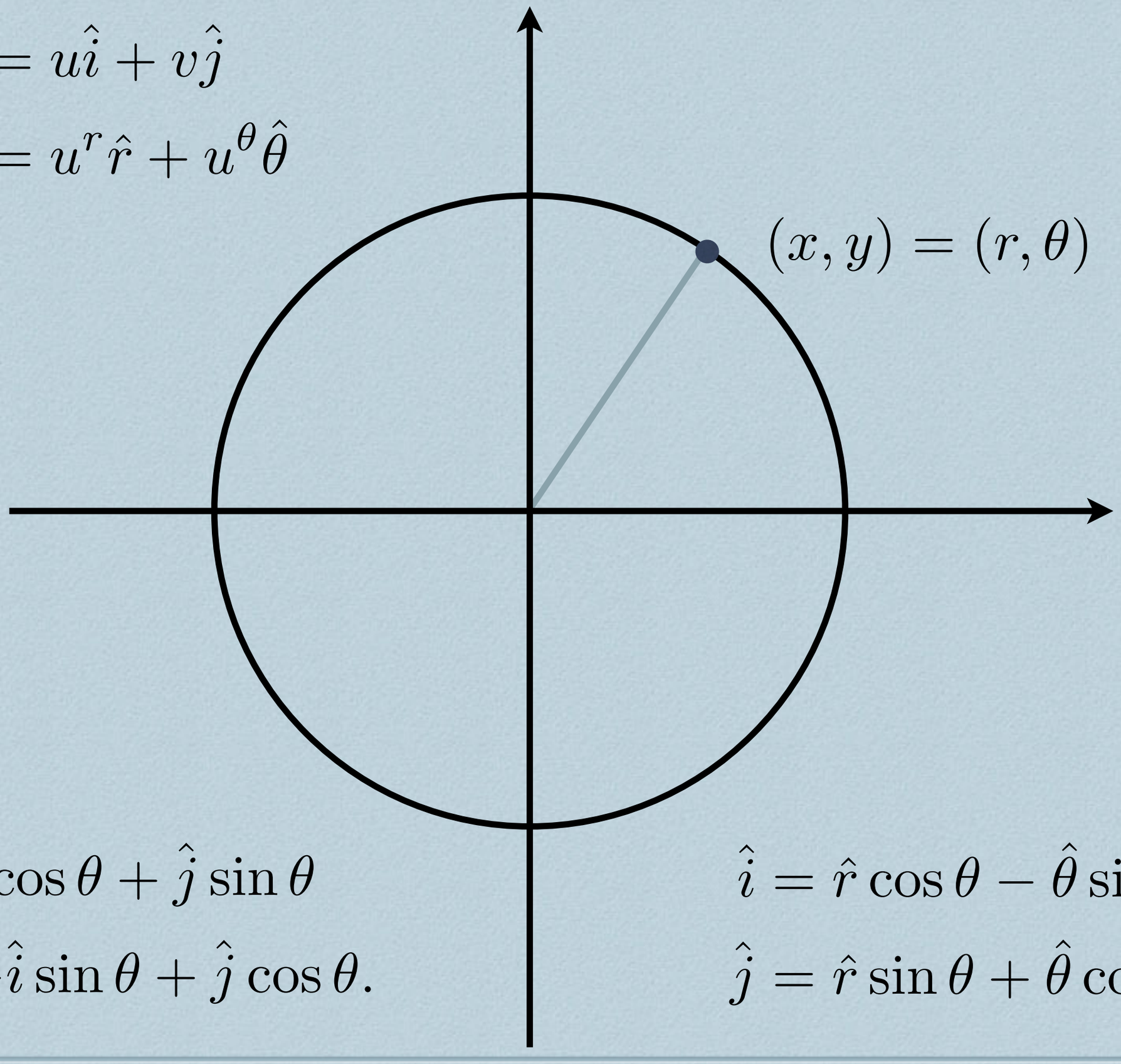
$$v_i = \text{ number } \hat{e}_i$$
$$\tau_{ij} = \text{ number } \hat{e}_i \otimes \hat{e}_j$$

- In practice the dyads are used to first write the vector gradient.

- The central difficulty is to not make mistakes while carrying out the partial derivatives of the components of the velocity and the basis vectors.

- However once we have the vector gradient the rest is just matrix algebra. So let's see how it works.

$$\vec{u} = u\hat{i} + v\hat{j}$$
$$= u^r\hat{r} + u^\theta\hat{\theta}$$



$$\hat{r} = \hat{i}\cos\theta + \hat{j}\sin\theta$$
$$\hat{\theta} = -\hat{i}\sin\theta + \hat{j}\cos\theta.$$

$$\hat{i} = \hat{r}\cos\theta - \hat{\theta}\sin\theta$$
$$\hat{j} = \hat{r}\sin\theta + \hat{\theta}\cos\theta.$$

$$\nabla = \partial_x \hat{i} + \partial_y \hat{j}$$

$$\vec{\nabla}\vec{u} = \partial_x u \hat{i}\hat{i} + \partial_y u \hat{i}\hat{j} + \partial_x v \hat{j}\hat{i} + \partial_y v \hat{j}\hat{j}$$

$$\begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix}$$

- For traditional Cartesian coordinates the notation seems almost unnecessary (though the matrix form is convenient).

- But using the outer product does allow us to make a link between mathematics and the physical interpretation of the stress and strain tensors.

- For any coordinates for which the axes vary with position (like the polar coordinates on the last slide) the notation is vital.

- Start by writing out the form of the gradient in the polar basis:

$$\vec{\nabla} = \hat{r}\partial_r + \hat{\theta}\frac{1}{r}\partial_\theta.$$

$$\partial_r \hat{r} = 0$$

$$\partial_\theta \hat{r} = \hat{\theta}$$

$$\partial_r \hat{\theta} = 0$$

$$\partial_\theta \hat{\theta} = -\hat{r}.$$

- First find the partial derivatives of the basis vectors as on left.
- Then apply the gradient operator in polar coordinates to the vector expressed in polar coordinates.
- Don't forget to use the product rule to take the partial derivatives of the components of the vector AND the basis vectors.

$$\vec{\nabla}\vec{u} = \left(\hat{r}\partial_r + \hat{\theta}\frac{1}{r}\partial_\theta\right)\left[u^r\hat{r} + u^\theta\hat{\theta}\right]$$

- The main effect of the spatially changing basis vectors is to make it vital to keep track of which dyad a particular partial derivative contributes to.

$$\vec{\nabla}\vec{u} = \hat{r}\hat{r}\partial_r u^r + \hat{\theta}\hat{r}(\tfrac{1}{r}\partial_\theta u^r - \tfrac{u^\theta}{r}) + \hat{r}\hat{\theta}\partial_r u^\theta + \hat{\theta}\hat{\theta}(\tfrac{1}{r}\partial_\theta u^\theta + \tfrac{u^r}{r})$$

- Care must be taken with the algebra, and the resulting expression is much neater in matrix form.

$$\vec{\nabla}\vec{u} = \begin{bmatrix} \partial_r u^r & (\frac{1}{r}\partial_\theta u^r - \frac{u^\theta}{r}) \\ \partial_r u^\theta & (\frac{1}{r}\partial_\theta u^\theta + \frac{u^r}{r}) \end{bmatrix}$$

- Once the vector gradient is available in matrix form the rate of strain and stress tensors can be found simply by matrix manipulations.

- The real issue comes about when we wish to find the divergence of the stress tensor in the Navier Stokes equations (books often use the misleading shorthand "vector Laplacian" for this term).

- Here is the expression of the divergence of the first row:

$$2\mu(\hat{r}\partial_r + \hat{\theta}\tfrac{1}{r}\partial_\theta) \cdot (\hat{r}\hat{r}\partial_r u^r + \hat{r}\hat{\theta}\left[\tfrac{r}{2}\partial_r\left(\frac{u^\theta}{r}\right) + \tfrac{1}{2r}\partial_\theta u^r\right]).$$

- The product rule will now involve the partial derivatives of the components as well as the partial derivative of the dyads!

- Historically the results for standard systems of coordinates (polar, cylindrical and spherical) were recorded in fluid mechanics textbooks (Kundu's book has a nice section as an Appendix).

- Computer algebra packages offer an appealing alternative (see Maple worksheets posted with these slides), though it can be a bit of work to get Maple to carry out physically meaningful calculations.

- The link below shows how one would derive the material derivative in polar coordinates.

- http://pleasemakeanote.blogspot.com/2008/08/derivation-of-navier-stokes-equations_17.html

Using dyads in practice

- Choose a coordinate system.
- Find the relevant equations in a book, and possibly double check against a second reference.
- Simplify for the particular situation you are studying (not much point using other coordinate systems if things don't simplify).
- Work with your simplified equations.